

A dataset of NDN traffic traces for the research community

Sankalpa Timilsina, Susmit Shannigrahi — *Tennessee Tech*
Davide Pesavento, Junxiao Shi, Lotfi Benmohamed — *NIST*

NDN Community Meeting 2023

Importance of traffic traces

- Reproducible results, e.g., in simulations
 - Enable apples-to-apples comparison of different protocols and designs
- To help design forwarding strategies, caching policies, etc.
- Performance measurements for optimization and tuning
- Network troubleshooting and debugging
- “Given NIST’s expertise in metrology, placing more focus and resources on measurement tools, *synthetic and real traffic trace generation and maintenance*, [...] would be a major contribution to the field” [1]

[1] Recommendation following a 2022 review of NIST’s Communication Technology Laboratory by a panel of experts assigned by the National Academies of Sciences, Engineering, and Medicine (NASEM)

How to produce an NDN traffic trace?

- “Convert” from other traces that contain hierarchical names, e.g., HTTP, NFS
 - Source traces are hard to obtain, harder to publish
 - Source traces can be obfuscated but must preserve the name hierarchy
 - Cannot capture the full semantics of NDN
 - May not represent how native NDN apps interact

How to produce an NDN traffic trace?

- “Convert” from other traces that contain hierarchical names, e.g., HTTP, NFS
 - Source traces are hard to obtain, harder to publish
 - Source traces can be obfuscated but must preserve the name hierarchy
 - Cannot capture the full semantics of NDN
 - May not represent how native NDN apps interact
- Synthesize
 - What parameters?
 - Chicken-and-egg problem

How to produce an NDN traffic trace?

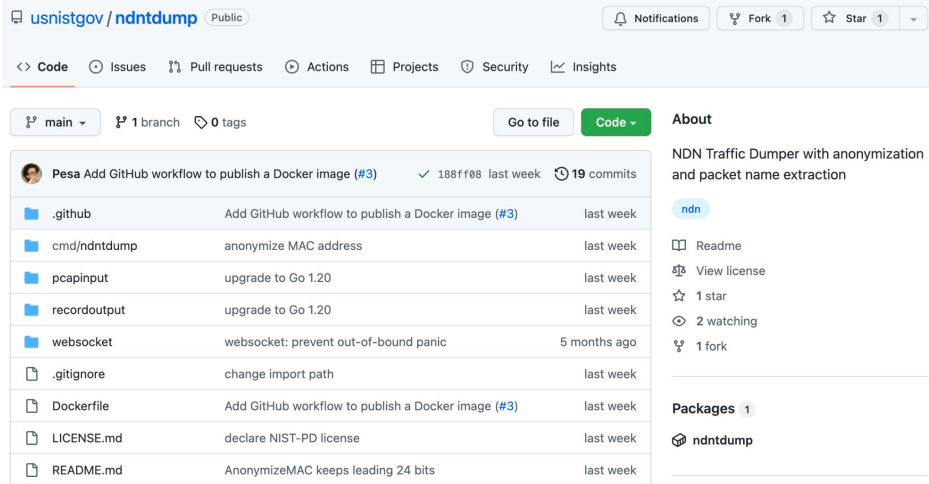
- “Convert” from other traces that contain hierarchical names, e.g., HTTP, NFS
 - Source traces are hard to obtain, harder to publish
 - Source traces can be obfuscated but must preserve the name hierarchy
 - Cannot capture the full semantics of NDN
 - May not represent how native NDN apps interact
- Synthesize
 - What parameters?
 - Chicken-and-egg problem
- Capture from a real/realistic NDN deployment
 - Not many NDN applications
 - Real users vs. synthetic users

Project goals

- **Dataset** of traffic traces
- Software **toolkit**
 - Automated traffic capture, anonymization, and compression
 - Replay the traces in ndnSIM and Mini-NDN
 - Analyze the traces to extract basic metrics and trends
 - Dissect the traces and take only a subset, e.g., only routing traffic
- Each piece is **modular** and can be used independently

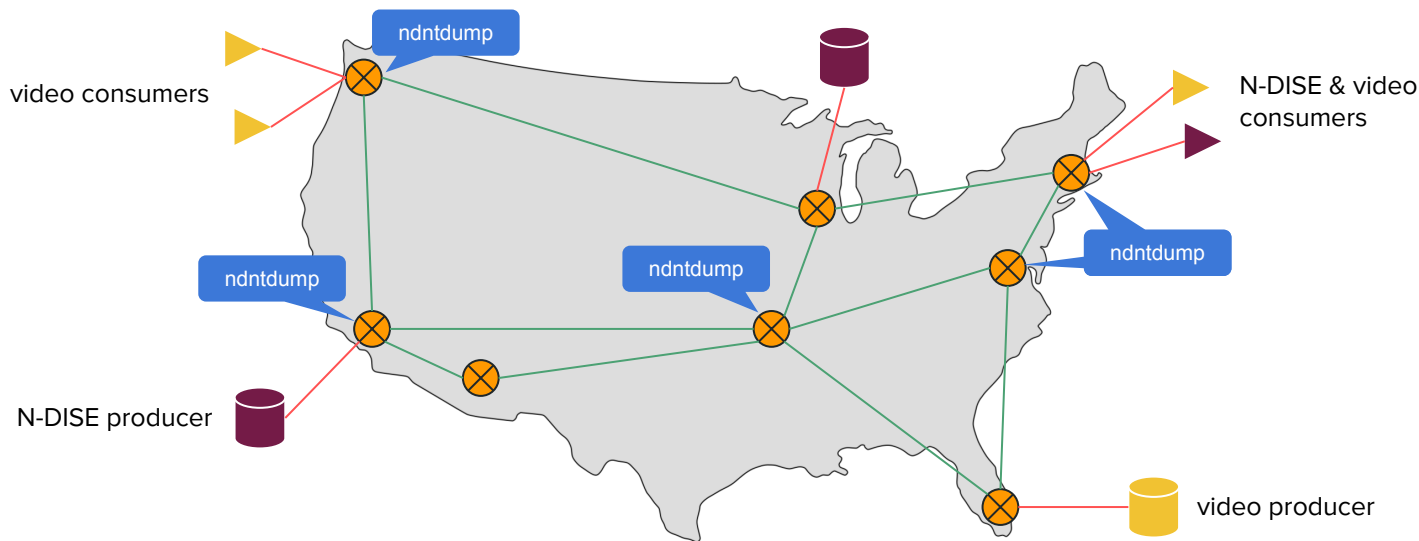
Traffic dumper

- First version already available
 - ◆ <https://github.com/usnistgov/ndntdump>
- Live capture from network interface
 - ◆ Or read an existing trace from file
- Save in *pcap* or *pcapng* format
 - ◆ With on-the-fly gzip or zstd compression
- Best-effort anonymization features
 - ◆ Replace Content and ApplicationParameters with zeros
 - Also increases the compression efficiency
 - ◆ LSBs of IPv4/IPv6/MAC addresses are XOR'd with a random key
 - ◆ Other prefix-preserving techniques are under investigation, e.g., Crypto-PAN



Trace collection on the NDN testbed

- Two applications for now
 - N-DISE, bulk transfer of large amounts of scientific data (CERN's LHC experiments)
 - Video streaming, based on Shaka Player and NDNts
- Almost ready to start collecting

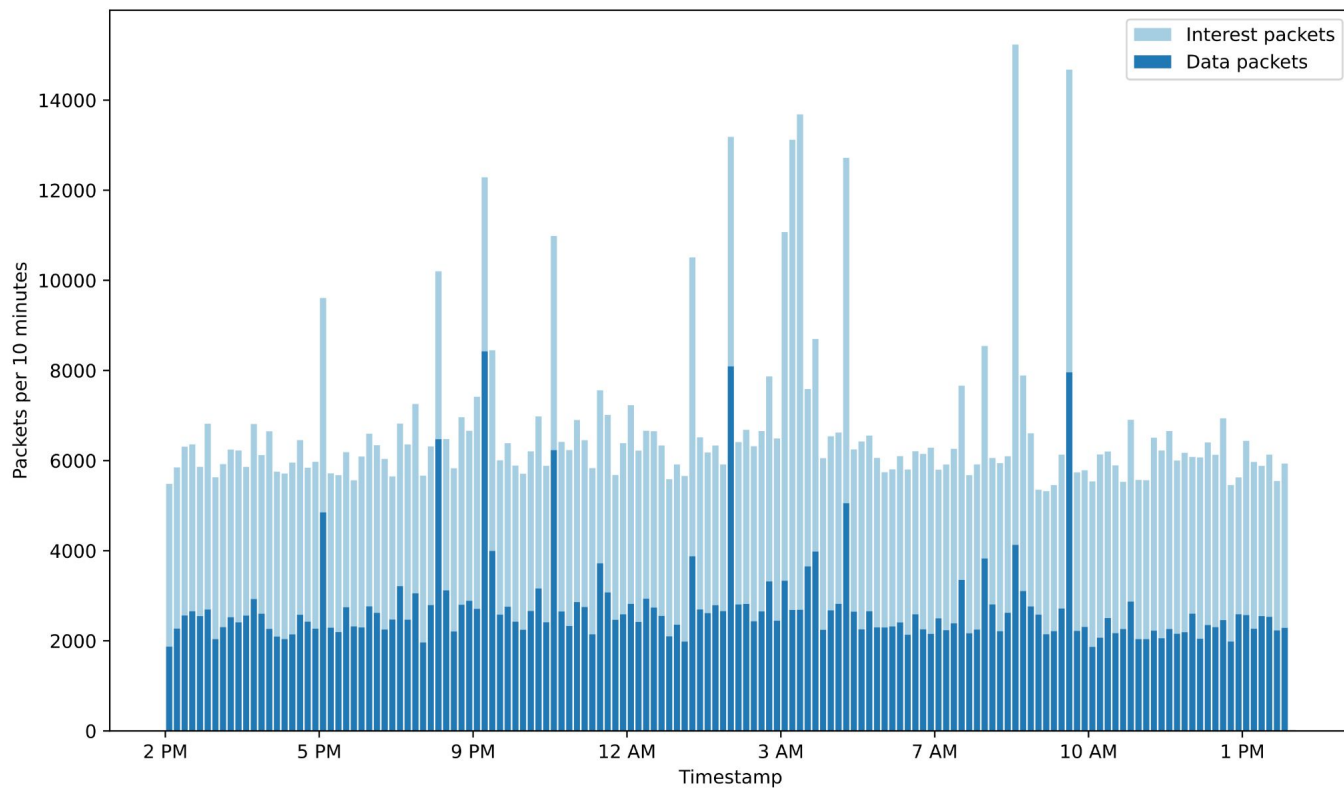


Trace collection on the NDN testbed

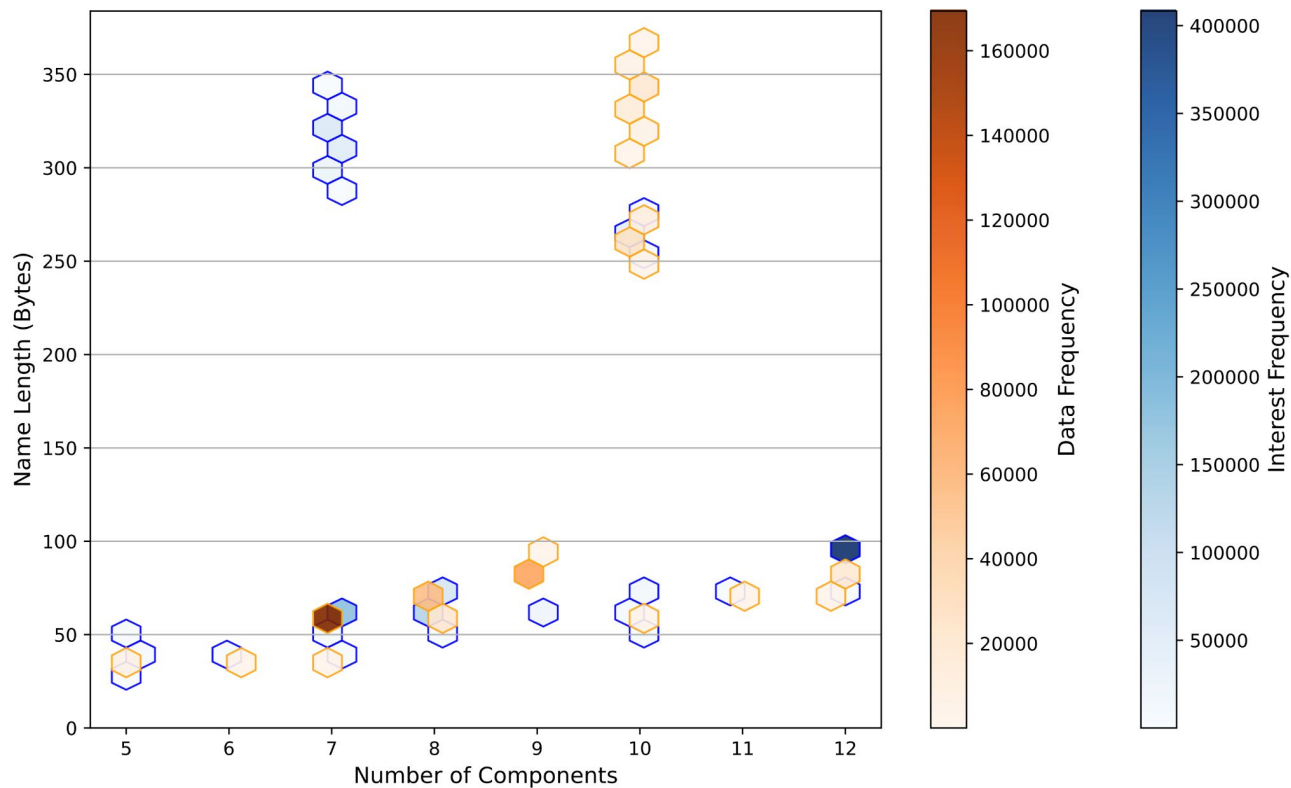
- Two applications for now
 - N-DISE, bulk transfer of large amounts of scientific data (CERN's LHC experiments)
 - Video streaming, based on Shaka Player and NDNts
- Almost ready to start collecting

- In the meantime, we are testing with a sample trace from Junxiao's ndn6 network
 - 24 hours, ~1.3 million packets
 - Traffic is mostly video streaming and NLSR

Sample trace – Number of packets per 10' interval



Sample trace – Name length vs. number of components



Conclusion and future directions

- Work in progress
- Seeking feedback
 - How useful is the dataset/toolkit?
 - Would you use it in your research? How?
 - How to make it more useful?
- Any other applications we can use to generate traffic?
- Different network environments (e.g., wireless)?
- In the future
 - Shared database with API?
 - Dashboard for real-time troubleshooting?
 - Integration with FABRIC?